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EIGSEP Battery Configuration and Setup

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1. INTRODUCTION

This document describes how to build, wire, and connect the main battery pack used for EIGSEP. The pack consists of two independent modules: a 4S (12V) module and a 2S (6V) module, each built from $LiFePO_4$ cells. Each module has its own Battery Management System (BMS) for cell protection and voltage balancing, its own inline fuse holder for overcurrent protection, and its own TX1 WH2 circuit breaker for manual disconnect. Together the two modules power the entire EIGSEP system and weigh approximately 50 lbs (∼23 kg).

A reader with no prior experience building battery packs should be able to follow this document from start to finish. Each section explains not just *what* to do, but *why*.

2. BACKGROUND

2.1. *$LiFePO_4$ cells*

$LiFePO_4$ (lithium iron phosphate) cells have a nominal voltage of 3.2 V each. Connecting cells in series (positive terminal of one cell to negative terminal of the next) adds their voltages. Four cells in series (4S) gives $4 \times 3.2 = 12.8$ V. Two cells in series (2S) gives $2 \times 3.2 = 6.4$ V.

2.2. *The Battery Management System (BMS)*

A BMS protects the cells from damage by monitoring each cell's voltage and cutting off current if any cell goes outside safe limits (overcharge, overdischarge, or overcurrent). It connects to the battery on two power terminals:

- `B-` — the main battery negative input to the BMS.
- `P-` — the protected negative output from the BMS to the load.

The BMS sits in the *negative* path only. The positive wire (`B+`) bypasses the BMS entirely and goes directly to the fuse and breaker. When the BMS trips, it opens the `P-` output, breaking the circuit and stopping current flow.

The BMS also has a separate low-current *balance port* (sometimes called the sampling connector). This multi-pin connector carries one wire per cell junction so the BMS can measure each cell's individual voltage. It carries no load current and must be connected before the main power wires.

2.3. *Fuse and circuit breaker*

Each module uses two layers of overcurrent protection on the positive line:

- **Inline fuse holder** — an ATO/ATC blade fuse holder wired close to the battery positive terminal. Each module has its own dedicated fuse holder. Protects the wiring between the battery and the breaker from a catastrophic short circuit. Sacrificial: the fuse must be replaced after blowing.
- **TX1 WH2 circuit breaker** — a hydraulic-magnetic breaker downstream of the fuse. Acts as a manual on/off switch and also trips automatically under sustained overcurrent. Resets by flipping the handle back on — no replacement needed.

2.4. *Output connector*

Each module's output is a single N-type coaxial connector. The center pin carries the positive supply (`B+` path), and the outer shield carries the negative return (`P-` path). One connector provides a complete circuit to the load.

3. ASSEMBLY

3.1. *Building the 4S (12V) module*

Connect four 3.2 V $LiFePO_4$ cells in series using shunts: the negative terminal of cell 1 connects to the positive terminal of cell 2, the negative of cell 2 to the positive of cell 3, and the negative of cell 3 to the positive of cell 4 (Figure 2). The main positive terminal of the pack (`B+`) is the free positive of cell 1. The main negative terminal (`B-`) is the free negative of cell 4, which connects to the BMS. Arrange the cells so both main terminals are accessible for wiring.

The 2S (6V) module is built the same way with two cells, giving a `B+` and `B-` terminal.

3.2. *Attaching the BMS*

Important: Connect the balance/sampling wires before connecting any power wires to the BMS. The BMS uses these to verify cell voltages before enabling its output.

Step 1 — Balance port (sampling wires). The balance connector has one wire per cell junction. Connect them in order from the most negative point of the pack to the most positive:

1. `B0` to the main negative of the pack (the negative terminal of cell 1).
2. `B1` to the junction between cell 1 and cell 2.
3. `B2` to the junction between cell 2 and cell 3.
4. `B3` to the junction between cell 3 and cell 4.
5. `B4` to the main positive of the pack (the positive terminal of cell 4).

For the 2S module, connect only `B0`, `B1`, and `B2`.

Step 2 — Main negative (`B-`). Run a heavy gauge wire from the main negative terminal of the battery pack (`B-`) to the `B-` terminal on the BMS. This is the negative power input to the BMS.

Step 3 — Do not connect `P-` yet. Leave the `P-` terminal disconnected until the rest of the circuit is wired.

3.3. *Wiring the positive path*

Run a heavy gauge wire from the main positive terminal of the battery pack (`B+`) to the input lead of the inline fuse holder. Insert a 20 A ATO blade fuse into the holder. From the output lead of the fuse holder, run a wire to one blade terminal of the TX1 WH2 circuit breaker. The other blade terminal of the TX1 WH2 is the positive output of the module — connect this to the center pin of the N-type output connector.

3.4. *Completing the negative path*

Run a wire from the `P-` terminal of the BMS to the shield of the N-type output connector. This completes the circuit: `B-` enters the BMS, and the protected `P-` output exits to the load via the connector shield.

The full circuit for each module is shown in Figure 1.

3.5. *Final check before connecting a load*

Before connecting any load, verify the following:

1. All balance wires are seated in the BMS balance port.
2. The `B-` wire is connected to the BMS `B-` terminal.
3. The `P-` wire is connected to the BMS `P-` terminal and to the N-type shield.
4. The positive path runs: battery `B+` → inline fuse holder → TX1 WH2 → N-type center pin.
5. The TX1 WH2 breaker is in the OFF position before connecting a load.
6. Measure the voltage between the N-type center pin and shield with a multimeter. It should read the full pack voltage: 12.8–14.6 V for the 4S module (nominal 12.8 V, fully charged 14.4–14.6 V) or 6.4–7.3 V for the 2S module (nominal 6.4 V, fully charged 7.2–7.3 V). If it reads zero with the breaker OFF, that is correct — flip the breaker ON to verify.

4. OPERATING THE SYSTEM

To power on a module, flip its TX1 WH2 breaker to the ON position. To power off, flip it to OFF. The two modules are completely independent and can be switched on or off separately.

If the BMS trips due to a fault (overcurrent, cell overvoltage, or cell undervoltage), the `P-` output opens and the load loses power. The breaker handle will remain in the ON position — this is not a breaker trip. Identify and resolve the fault, then power-cycle the BMS by briefly disconnecting and reconnecting `B-`.

5. FIGURES

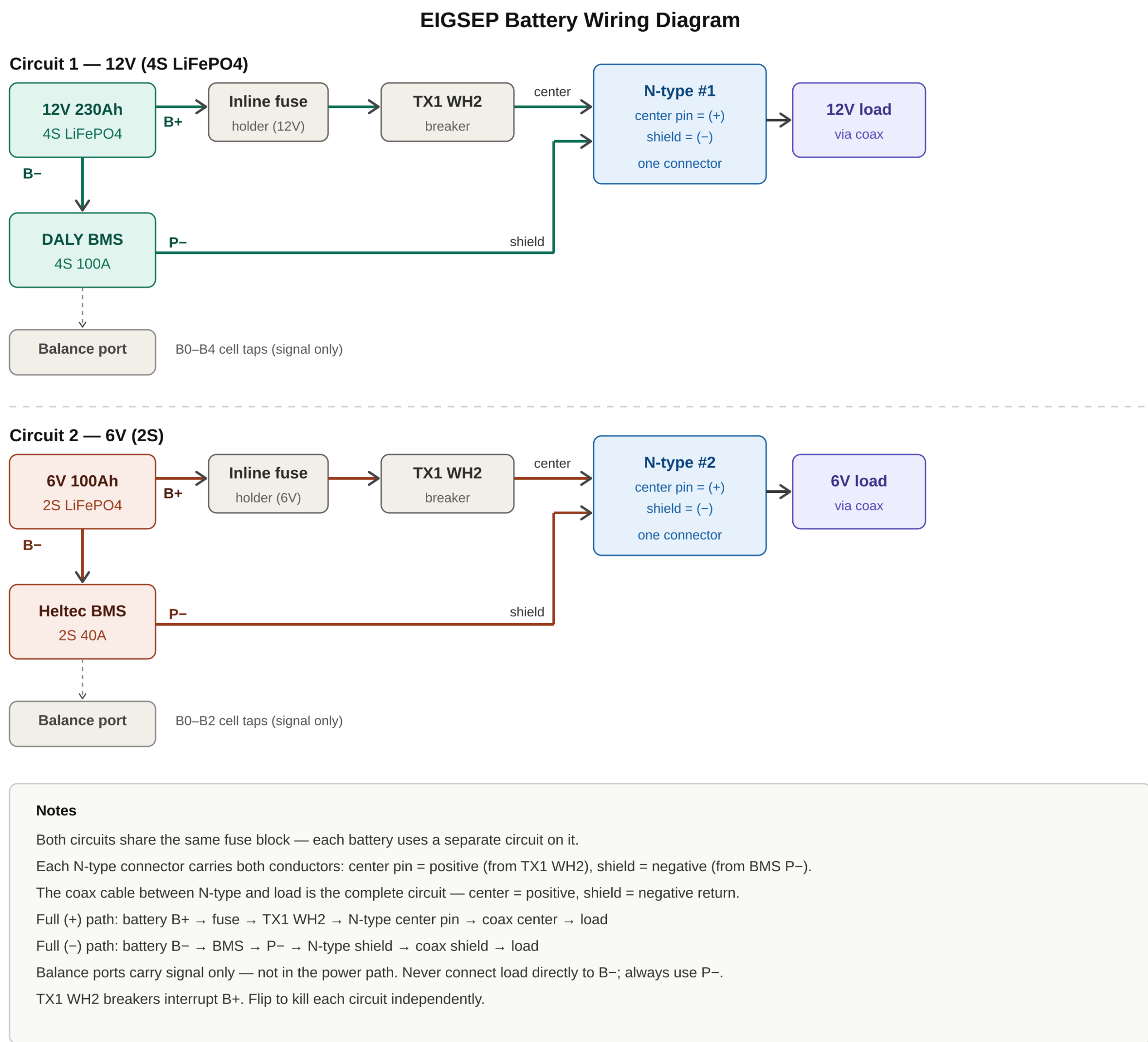

Figure 1. Full wiring schematic for both the 12V (4S) and 6V (2S) circuits. Each circuit is independent and has its own inline fuse holder. Positive path (green/red): battery `B+` → inline fuse holder → TX1 WH2 breaker → N-type center pin → load. Negative path: battery `B-` → BMS → `P-` → N-type shield → load. The balance port (dashed line) carries cell-voltage signals only and is not part of the power path.

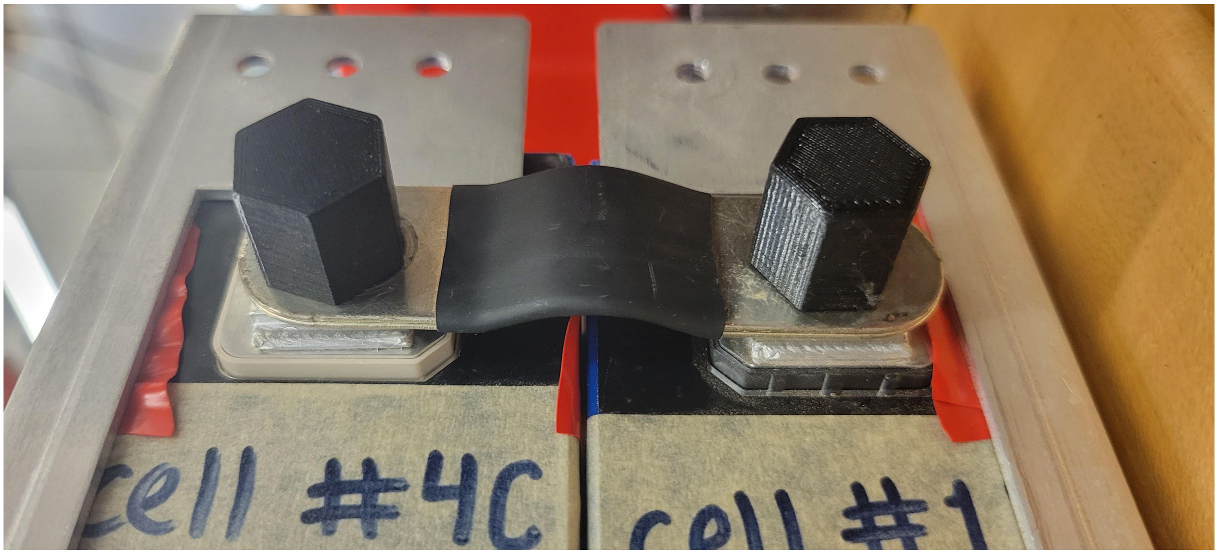

Figure 2. Two $LiFePO_4$ cells connected in series using a shunt. The negative terminal of the lower cell connects to the positive terminal of the upper cell. Repeating this connection across all cells in the module produces the full pack voltage.

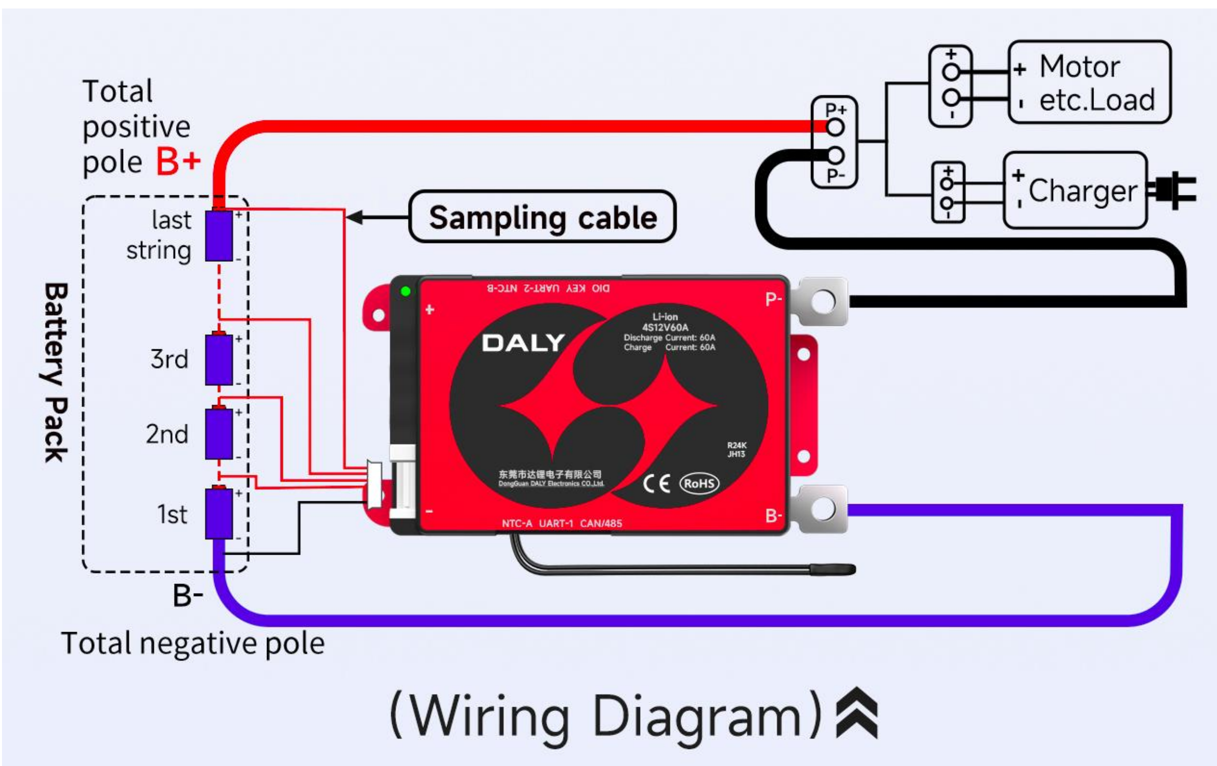

Figure 3. DALY BMS wiring diagram for the 4S (12V) module, showing the balance port connections (B0–B4) and the main power terminals (`B-` input and `P-` output). The 2S module BMS connects identically but uses only B0–B2.

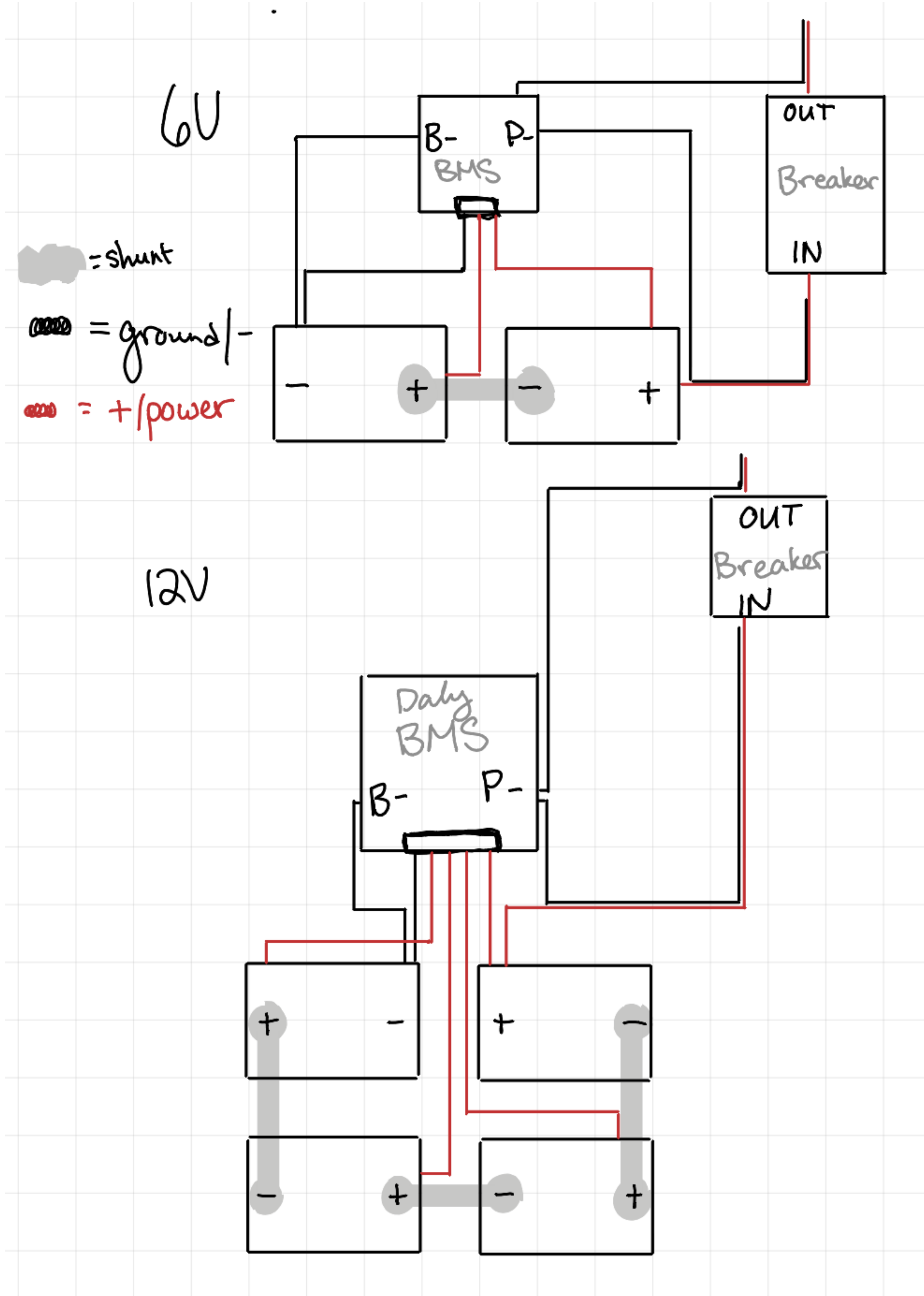

Figure 4. Physical layout of both modules showing cell arrangement, BMS placement, and terminal locations for the 12V (4S) and 6V (2S) packs.

REFERENCES

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https://www.dalybms.com/4s-bms-wiring-tutorial/